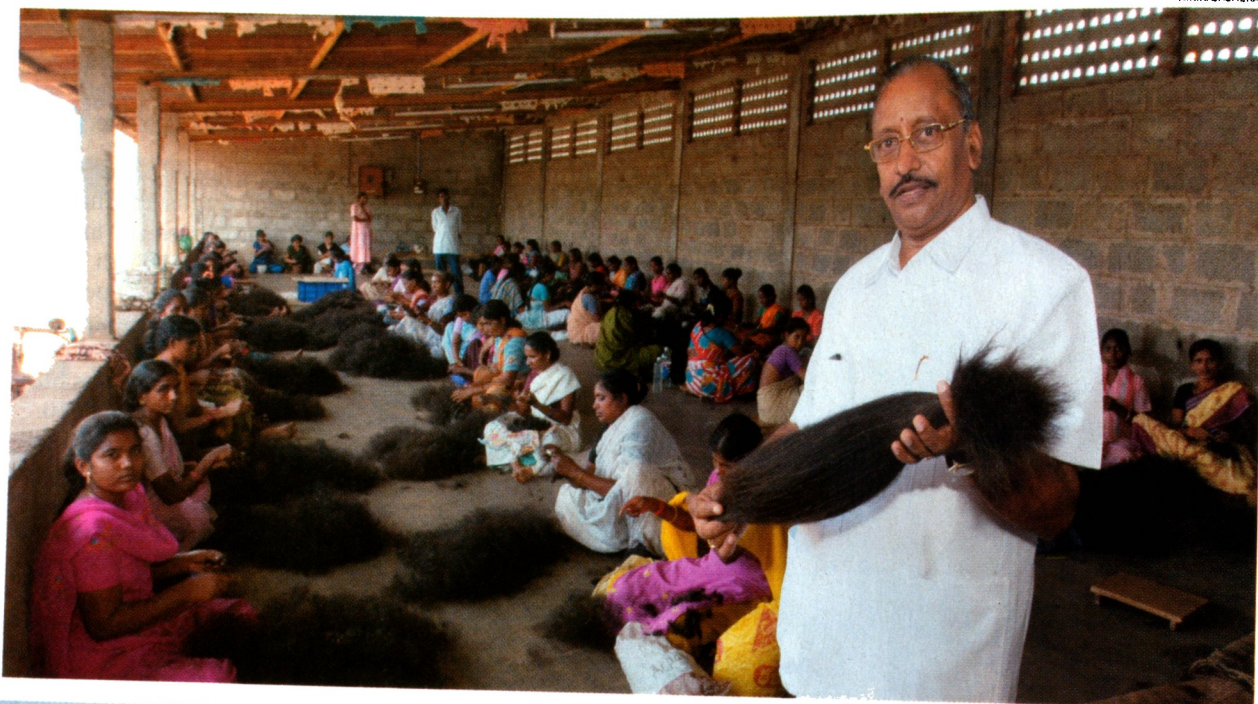




▲ **M.M. GUPTA, CHAIRMAN, GUPTA GROUP**
His group has been processing and exporting human hair since 1974 and has a turnover of Rs 225 crore.



RAVI S SAHANI

▲ **MANIK THAPAR, CEO, ECO WISE**
Provides waste management solutions by treating municipal solid wastes and now has a turnover of Rs 1.2 crore.

whom deposit it at Gupta's collection centres in Andhra Pradesh and Karnataka. Hair in its raw form commands Rs 1,700 a kilo.

Entrepreneurs are increasingly seeking ways to turn rubbish into value. Mumbai startup, Sustainable Technologies and Environmental Projects (STEPS), set up in June 2007 by T. Raghavendra Rao, uses a thermal catalytic conversion method called 'polycrack' to convert plastic and organic waste like kitchen, animal and agro refuse into petroleum fuels like

gas, liquid fuel or a combination of both. A kg of polypropylene plastic feedstock yields 1.2 litre of liquid fuel that can be used to power electrical generators, or 150g of LPG, at a conversion cost of Rs 11 per litre. Although the industry has a record of failure, Rao is undeterred. He argues that the technology is "far superior to

traditional recyclers, which require clean plastics or involve high costs". With rising oil prices, he may just have timed it right.

Profits may still elude, but optimism and innovation are in plentiful supply. If the chemical engineering department of Jadavpur University, Kolkata, is collaborating with Moromi, an NGO, to extract dyes from waste flowers for use in the textile industry as well as for bio-fertilisers, Rajendra Gandhi, MD of the Mumbai-based Rs 110-crore Gujarat Reclaim and Rubber Products has been successfully recycling old rubber from tyres since the '70s. With natural rubber in short supply, tyre companies, ranging from Goodyear, Michelin and Bridgestone abroad to Apollo and JK Tyres in India, mix 5-6 per cent reclaimed rubber in car tyres, whereas bicycle tyres contain about 40 per cent. Says Gandhi, "It is no longer enough to manage waste and minimise the environmental impact of its treatment. Waste must be used as a resource." Yet, rising manpower costs are forcing many of these players to rethink their strategies. The garbage warriors must plod on through millions of dustbins before they can have a good, clean run. ■